

■ 1.6.2 Functions as Procedures

In many kinds of calculations, you may find yourself typing the same input to *Mathematica* over and over again. You can save yourself a lot of typing by defining a *function* that contains your input commands.

This constructs a product of three terms, and expands out the result.

```
In[1]:= Expand[ Product[x + i, {i, 3}] ]
```

```
Out[1]= 6 + 11 x + 6 x2 + x3
```

This does the same thing, but with four terms.

```
In[2]:= Expand[ Product[x + i, {i, 4}] ]
```

```
Out[2]= 24 + 50 x + 35 x2 + 10 x3 + x4
```

This defines a function `exprod` which constructs a product of n terms, then expands it out.

```
In[3]:= exprod[n_] :=  
        Expand[ Product[ x + i, {i, 1, n} ] ]
```

Every time you use the function, it will execute the `Product` and `Expand` operations.

```
In[4]:= exprod[5]
```

```
Out[4]= 120 + 274 x + 225 x2 + 85 x3 + 15 x4 + x5
```

The functions you define in *Mathematica* are essentially “procedures” that execute the commands you give. You can have several steps in your procedures, separated by semicolons. Section 2.5 describes the full procedural programming capabilities of *Mathematica*.

The result you get from the whole function is simply the last expression in the procedure. Notice that you have to put parentheses around the procedure when you define it.

```
In[5]:= cex[n_, i_] := ( t = exprod[n];  
                        Coefficient[t, xi] )
```

This “runs” the procedure.

```
In[6]:= cex[5, 3]
```

```
Out[6]= 85
```